



Features

- Configurable overcurrent protection
- Overtemperature protection
- Open load detection
- Short circuit to GND detection
- Electrostatic Discharge (ESD) protection
- 16-Bit SPI (for diagnostic and control)
- Soldering: Automated Optical Inspection capability (AOI)
- Green product (completely lead free)
- AEC-Q100 qualified

Description

The HE9104C is a quad channel low-side with open drain DMOS output stages manufactured in Smart Power Technology. The HE9104C is protected by embedded protection functions and designed for automotive powertrain applications. The output stages can be controlled directly by parallel inputs for PWM applications (for example gasoline multiport injection) or by SPI.

Part Number	Package	Marking	Packing
HE9104CRGAA	ETSSOP20	HE9104C	3000/Tape&Reel

*Not sure if you got the latest datasheet or not. Please kindly contact the sales of KKChips to confirm. Thanks.

*The full part number will be confirmed after the official version of the product.

1 Overview

Table 1-1 Product summary

Symbol	Parameter	Min	Typ	Max	Unit
V_{IO}	Signal supply voltage	3.0	-	5.5	V
	Analog supply voltage	4.5	-	5.5	V
$V_{DS(AZ)}$	Output clamping voltage	50	-	60	V
$R_{DS(ON)}$	Typical On-state resistance CH 1-4 at $T_j = 25^{\circ}\text{C}$	-	150	-	m Ω
$R_{DS(ON)}$	Typical On-state resistance CH 1-4 at $T_j = 150^{\circ}\text{C}$	-	300	-	m Ω
I_D	Nominal load current CH 1-4 (continuous)	-	3	-	A
I_{SCB}	Short circuit to battery detection threshold CH 1-4	-	5	-	A

Table of contents

1	Overview	2
2	Block diagram	6
3	Pin configuration	7
3.1	Pin assignment.....	7
3.2	Pin definitions and functions	8
4	General product characteristics	9
4.1	Absolute maximum ratings	9
4.2	Operating conditions	10
5	Electrical and functional description of blocks	11
5.1	Power supply	11
5.2	Parallel inputs	12
5.3	Power stages	12
5.4	Protection functions.....	14
5.4.1	Overtemperature protection	14
5.4.2	Short circuit to battery protection	14
5.4.3	Overcurrent protections	14
5.5	Diagnostic functions	14
5.5.1	Output stage status	19
5.6	Communication watchdog	19
6	16 bit SPI interface	20
6.1	Electrical characteristics 16 bit SPI interface.....	20
6.2	SPI registers.....	22
6.2.1	Registers	24
7	Package outlines	34
8	Application information	36
9	Revision history	37

List of Figures

Figure 2-1	Block diagram.....	6
Figure 3-1	Pin configuration (top view).....	7
Figure 5-1	Overcurrent and short circuit to battery protection	15
Figure 5-2	Diagnostic functions (overview only).....	16
Figure 5-3	SCG and OL diagnostic function (overview only).....	16
Figure 6-1	SPI timing	20
Figure 7-1	ETSSOP20 package dimension.....	34
Figure 8-1	Multi port injection application diagram	36

List of Tables

Table 1-1	Product summary.....	2
Table 3-1	Pin configuration.....	8
Table 4-1	Absolute maximum ratings.....	9
Table 4-2	Operating conditions	10
Table 4-3	Thermal resistance.....	10
Table 4-4	Temperature range	10
Table 5-1	Power supply electrical characteristics	11
Table 5-2	Parallel inputs electrical characteristics	12
Table 5-3	Power outputs electrical characteristics	12
Table 5-4	Electrical characteristics: power outputs (cont'd)	13
Table 5-5	Diagnostic functions electrical characteristics	17
Table 5-6	Communication watchdog timeout configuration.....	19
Table 6-1	16 bit SPI interface electrical characteristics	20
Table 6-2	Register Address Space.....	23
Table 6-3	Register Overview	23
Table 6-4	Register Overview	23
Table 7-1	ETSSOP20L mechanical data	34
Table 9-1	Revision history.....	37

2 Block diagram

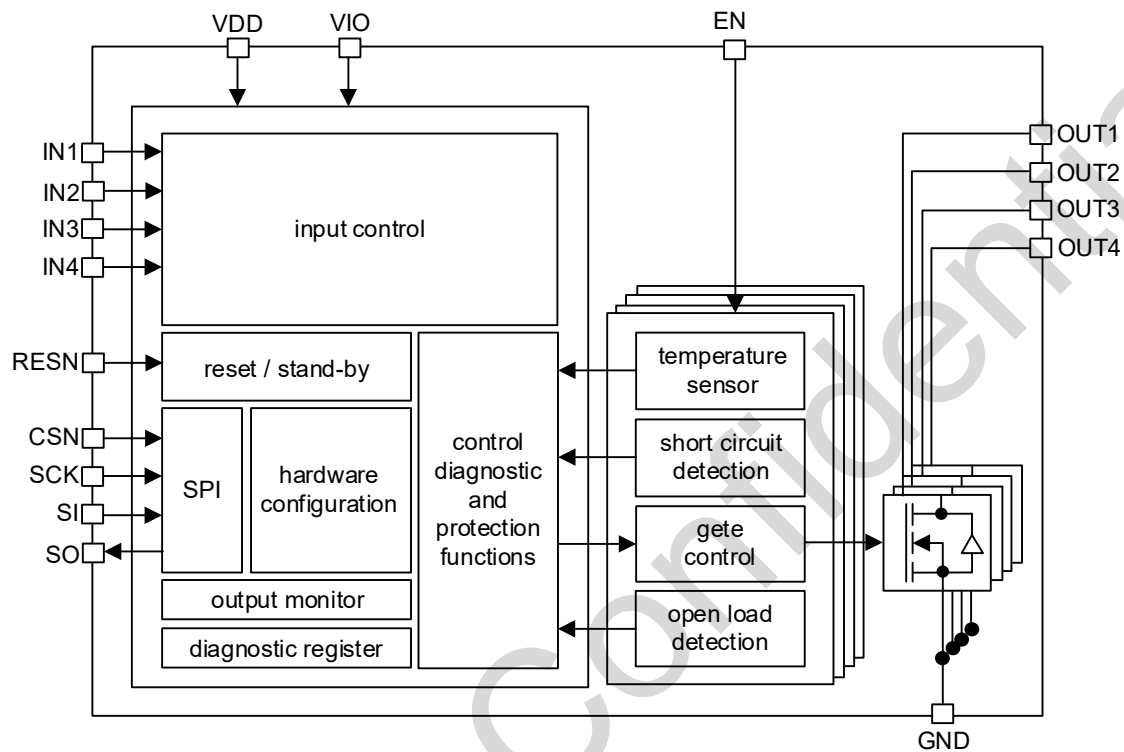


Figure 2-1 Block diagram

3 Pin configuration

3.1 Pin assignment

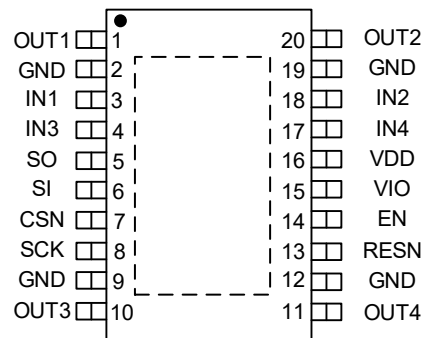


Figure 3-1 Pin configuration (top view)

3.2 Pin definitions and functions

Table 3-1 Pin configuration

#	Pin Name	Function
1	OUT1	Power Output 1
2	GND	Ground
3	IN1	Input 1
4	IN3	Input 3
5	SO	Serial Data Output
6	SI	Serial Data Input
7	CSN	Serial Chip Select (active low)
8	SCK	Serial Clock
9	GND	Ground
10	OUT3	Power Output 3
11	OUT4	Power Output 4
12	GND	Ground
13	RESN	Reset (active low)
14	EN	Output Enable
15	VIO	Signal Supply Voltage
16	VDD	Analog Supply Voltage
17	IN4	Input 4
18	IN2	Input 2
19	GND	Ground
20	OUT2	Power Output 2

Notes

1. The exposed pad of HE9104C is not connected to ground internally. It is highly recommended to connect the exposed pad to GND pins externally.
2. Pins 2 and 19 are the ground pins of outputs 1 and 2 and pins 9 and 12 are the ground pins of outputs 3 and 4. It is highly recommended to connect all GND pins externally.

4 General product characteristics

4.1 Absolute maximum ratings

$T_j = -40^{\circ}\text{C}$ to $+150^{\circ}\text{C}$, all voltages with respect to ground, positive current flowing into pin (unless otherwise specified)

Table 4-1 Absolute maximum ratings

Symbol	Parameter	Condition	Min.	Typ.	Max.	Unit	Param Num
V_{IO}	Signal supply voltage	–	-0.3	–	5.5	V	P_4.1.1
V_{DD}	Analog supply voltage	–	-0.3	–	5.5	V	P_4.1.2
V_{DS}	Continuous drain source voltage (OUT1 to OUT4)	–	-0.3	–	50	V	P_4.1.3
V_{IN}	Input voltage, all inputs and data outputs, sense lines	–	-0.3	–	$V_{IO}+0.3$	V	P_4.1.4
I_D	Output current per channel ¹⁾	Output ON	0	–	5.6	A	P_4.1.5
$V_{SC, \text{single}}$	Maximum voltage for short circuit protection (single event) ²⁾	–	–	–	30	V	P_4.1.6
V_{ESD1}	Electrostatic Discharge voltage - HBM (human body model) ³⁾	–	-2000	–	2000	V	P_4.1.7
V_{ESD2}	Electrostatic Discharge voltage - CDM (charge device model) ⁴⁾	–	-500	–	500	V	P_4.1.8

1) Output current rating as long as maximum junction temperature is not exceeded. The maximum output current in the application must be calculated using R_{thJA} depending on mounting conditions.

2) Short circuit is designed to be short circuit robust according to AEC-Q100-012.

3) According to ANSI/ESDA/JEDEC JS-001.

4) According to JESD22-C101.

Notes

- Stresses above the ones listed here may cause permanent damage to the device. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.
- Integrated protection functions are designed to prevent IC destruction under fault conditions described in the datasheet. Fault conditions are considered as “outside” normal operating range. Protection functions are not designed for continuous repetitive operation.

4.2 Operating conditions

Table 4-2 Operating conditions

Symbol	Parameter	Condition	Min.	Typ.	Max.	Unit	Param Num
E_{AR}	Maximum output clamping energy, linearly decreasing current ^{1) 2)}	$I_{D(0)} = 1.4 \text{ A}$, $T_{J(0)} = 110^\circ\text{C}$, Cycles: 1 billion	14	–	–	mJ	P_4.2.6
E_{AS}	Maximum output clamping energy, linearly decreasing current ³⁾	$T_J = 85^\circ\text{C}$, Cycles: 10	35	–	–	mJ	P_4.2.13
E_{AS}	Maximum output clamping energy, linearly decreasing current	$T_J = 145^\circ\text{C}$, Cycles: 10	25	–	–	mJ	P_4.2.14
$E_{AR,p}$	Maximum output clamping energy in parallel mode	OUT1&2 or OUT3&4, $I_{D(0), P} = 1.8 \times I_{D(0)}$	$1.7 \times E_{AR}$	–	–	mJ	P_4.2.2

Table 4-3 Thermal resistance

Symbol	Parameter	Condition	Min.	Typ.	Max.	Unit	Param Num
R_{thJA}	Junction to ambient	PV = 3 W, homogenously distributed between all output stages	–	14.47	–	K/W	P_4.2.3

Table 4-4 Temperature range

Symbol	Parameter	Condition	Min.	Typ.	Max.	Unit	Param Num
T_J	Operating temperature range	–	-40	–	150	°C	P_4.2.4
T_{stg}	Storage temperature range	–	-55	–	150	°C	P_4.2.5

1) Pulse shape represents inductive switch off: $I_D(t) = I_{D(0)} \times (1 - t/t_{pulse})$; $0 < t < t_{pulse}$

2) The given energy values are based on a cumulative scenario as specified in the Notes column.

3) The given energy values are based on a cumulative scenario as specified in the Notes column.

Note:

Within the functional range, the IC operates as described in the circuit description. The electrical characteristics are specified within the conditions given by the related electrical characteristics table.

All parameters are tested in the temperature range T_J -40 - 150°C; device functionality at high temperature is guaranteed by bench validation, electrical parameters are guaranteed by correlation with ATE tests at reduced temperature and adjusted limits (if needed).

5 Electrical and functional description of blocks

5.1 Power supply

The HE9104C is supplied by analog power supply line V_{DD} and signal power supply V_{IO} . A capacitor between pins V_{DD} to GND and V_{IO} to GND is recommended. After start-up of the power supply, the RESN pin should be kept low until the Reset Duration Time has expired. This will reset all SPI configuration registers and OUTx_STAT to their default values. In order to enable the output stages the EN pin has to be kept high and OUT_EN register has to be set.

V_{DD} = 4.5 V to 5.5 V, V_{IO} = 3 V to 5.5 V, T_j = -40°C to +150°C, (unless otherwise specified)

all voltages with respect to ground, positive current flowing into pin.

Table 5-1 Power supply electrical characteristics

Symbol	Parameter	Condition	Min	Typ	Max	Unit	Param Num
V_{IO}	Signal supply voltage	–	3	–	5.5	V	P_5.1.1
V_{DD}	Analog supply voltage	–	4.5	–	5.5	V	P_5.1.2
$I_{DD(on)}$	Supply current	–	–	–	10	mA	P_5.1.3
$V_{RESN(L)}$	Input low voltage of pin RESN	–	-0.3	–	1	V	P_5.1.4
$V_{RESN(H)}$	Input high voltage of pin RESN	–	2	–	$V_{IO} + 0.3$	V	P_5.1.5
$V_{RESN(Hys)}$	Hysteresis voltage of pin RESN	–	100	300	500	mV	P_5.1.6
I_{RESN}	Input pull-up current through pin RESN	$V_{RESET} = 0$ V	-100	-65	-30	μA	P_5.1.7
$t_{RESN(L)}$	Reset duration time ¹⁾	–	10	–	–	μs	P_5.1.8
$V_{EN(L)}$	Input low voltage of pin EN	–	-0.3	–	1	V	P_5.1.9
$V_{EN(H)}$	Input high voltage of pin EN	–	2	–	$V_{IO} + 0.3$	V	P_5.1.10
$V_{EN(Hys)}$	Hysteresis voltage of pin EN	–	100	300	500	mV	P_5.1.11
I_{EN}	Input pull-up current through pin EN	$V_{EN} = 0$ V	-100	-65	-30	μA	P_5.1.12

1) For proper startup, after the supply V_{DD} has reached its final voltage, the RESN pin should be held low until the reset duration time has expired.

5.2 Parallel inputs

Each input signal controls the output stage of its related channel. For example, IN1 controls OUT1, IN2 controls OUT2 etc. Input signals are active low. Hence, applying a voltage less than $V_{IN(L)}$ to INx turns OUTx on. It is possible to connect OUT1-2 and OUT3-4 in parallel. For this purpose the right configuration has to be selected in the CFG register. In this case IN1 controls OUT1-2 and IN3 controls OUT3-4.

$V_{DD} = 4.5\text{ V to }5.5\text{ V}$, $V_{IO} = 3\text{ V to }5.5\text{ V}$, $T_J = -40^\circ\text{C to }+150^\circ\text{C}$, (unless otherwise specified)
all voltages with respect to ground, positive current flowing into pin (unless for pin SO)

Table 5-2 Parallel inputs electrical characteristics

Symbol	Parameter	Condition	Min	Typ	Max	Unit	Param Num
$V_{IN(L)}$	Input low voltage of pin INx	–	-0.3	–	1	V	P_5.2.1
$V_{IN(H)}$	Input high voltage of pin INx	–	2	–	$V_{IO} + 0.3$	V	P_5.2.2
$V_{IN(Hys)}$	Input voltage hysteresis	–	100	300	500	mV	P_5.2.3
$I_{IN(L)}$	Input pull-up current through pin INx	$V_{IN} = 0\text{ V}$	-100	-65	-30	μA	P_5.2.4

5.3 Power stages

$V_{DD} = 4.5\text{ V to }5.5\text{ V}$, $V_{IO} = 3\text{ V to }5.5\text{ V}$, $T_J = -40^\circ\text{C to }+150^\circ\text{C}$, (unless otherwise specified)
all voltages with respect to ground, positive current flowing into pin (unless for pin SO)

Table 5-3 Power outputs electrical characteristics

Symbol	Parameter	Condition	Min.	Typ.	Max.	Unit	Param Num
$R_{DS(ON)}$	ON resistance	$T_J = 25^\circ\text{C}$,	–	150		m Ω	P_5.3.1
$R_{DS(ON)}$	ON resistance	$T_J = 150^\circ\text{C}$,	–	300	350	m Ω	P_5.3.2
$R_{DS(ON)}$	ON resistance in parallel mode	$T_J = 25^\circ\text{C}$, outputs 1&2 or 3&4 in parallel	–	75	–	m Ω	P_5.3.3
$R_{DS(ON)}$	ON resistance in parallel mode	$T_J = 150^\circ\text{C}$, outputs 1&2 or 3&4 in parallel	–	150	175	m Ω	P_5.3.4

$V_{DD} = 4.5 \text{ V to } 5.5 \text{ V}$, $V_{IO} = 3 \text{ V to } 5.5 \text{ V}$, $T_j = -40^\circ\text{C to } +150^\circ\text{C}$, (unless otherwise specified)
all voltages with respect to ground, positive current flowing into pin (unless for pin SO)

Table 5-4 Electrical characteristics: power outputs (cont'd)

Symbol	Parameter	Condition	Min	Typ	Max	Unit	Param Num
$V_{DS(AZ)}$	Output clamping voltage	output OFF	50	–	60	V	P_5.3.5
$I_{D(lkg)}$	Output leakage current	RESN=0, $T_{max} = 150^\circ\text{C}$	–	–	10	μA	P_5.3.6
I_{OUTx_OFF}	Output off-state current	RESN=1, OUTx_DIAG_EN = 0, $V_{OUTx} = 35 \text{ V}$	–	–	30	μA	P_5.3.7
t_{ON}	Turn-on time @ CFG1.SR = 00 _b	from 50% of INx to 20% of Vbat	–	15	–	μs	P_5.3.8
	Turn-on time @ CFG1.SR = 01 _b	from 50% of INx to 20% of Vbat	–	10	–	μs	
	Turn-on time @ CFG1.SR = 10 _b	from 50% of INx to 20% of Vbat	–	5	–	μs	
	Turn-on time @ CFG1.SR = 11 _b	from 50% of INx to 20% of Vbat	–	2.5	–	μs	
t_{OFF}	Turn-off time @ CFG1.SR = 00 _b	from 50% of INx to 80% of Vbat	–	15	–	μs	P_5.3.9
	Turn-off time @ CFG1.SR = 01 _b	from 50% of INx to 80% of Vbat	–	10	–	μs	
	Turn-off time @ CFG1.SR = 10 _b	from 50% of INx to 80% of Vbat	–	5	–	μs	
	Turn-off time @ CFG1.SR = 11 _b	from 50% of INx to 80% of Vbat	–	2.5	–	μs	
$T_{J(OT)}$	Overtemperature shutdown threshold	–	175	–	200	$^\circ\text{C}$	P_5.3.10

5.4 Protection functions

The HE9104C provides embedded protection functions. Integrated protection functions are designed to prevent IC destruction under fault conditions. Fault conditions are considered “outside” the normal operating range. Protection functions are not designed for continuous repetitive operation. Following protection functions are implemented for HE9104C:

- Overtemperature protection (OT).
- Short circuit to battery protection (SCB).
- Overcurrent protection (OC).

Note:

Integrated protection functions are designed to prevent IC destruction under fault conditions described in the datasheet. Fault conditions are considered as “outside” normal operating range. Protection functions are not designed for continuous repetitive operation.

5.4.1 Overtemperature protection

A dedicated temperature sensor for each channel detects if the temperature of its channel exceeds the overtemperature shutdown threshold. If the channel temperature exceeds the overtemperature shutdown threshold, the overheated channel is switched off immediately to prevent destruction. The channel can be turned on again after clearing the overtemperature error; however, if the sensed temperature is still higher than the overtemperature shutdown threshold the channel will switch off after the filter time t_{OT} .

5.4.2 Short circuit to battery protection

The HE9104C is protected in case of short circuit to battery. If the current of an output channel exceeds I_{SCB} , the respective channel is switched off immediately. The channel can be turned on again after the fault condition has been removed and the error has been cleared.

5.4.3 Overcurrent protections

The HE9104C is protected with configurable overcurrent protection. If the current of an output channel exceeds I_{OC} , the respective channel is switched off after the filter time $t_{d(OC)}$. The channel can be turned on again after the fault condition has been removed and the error has been cleared. Both current limit threshold I_{OC} and its filter time $t_{d(OC)}$ are configurable via SPI. The filter time, $t_{d(OC)}$, can only be configured while the output bit, OUT_EN , is low in the SPI register.

5.5 Diagnostic functions

Following diagnosis functions are implemented for all output stages of HE9104C:

- Short to battery detection (SCB) can be detected if stages are turned on.
- Overtemperature detection (OT) can be detected if stages are turned on.
- Time based overcurrent detection (OCF) can be detected if stages are turned on.
- Temperature based overcurrent detection (OCT) can be detected if stages are turned on.
- Short to GND detection (SCG) can be detected if stages are turned off.
- Open load detection (OL) can be detected if stages are turned off.

The diagnosis information of HE9104C can be accessed via SPI interface. OL and SCG diagnosis are recognized using two thresholds ($V_{OUTn-SCG}$ and $V_{OUTn-OL}$). It is also possible to turn off the internal diagnostic pull-down and pull-up current sources. In this case diagnosis of OL and SCG are deactivated.

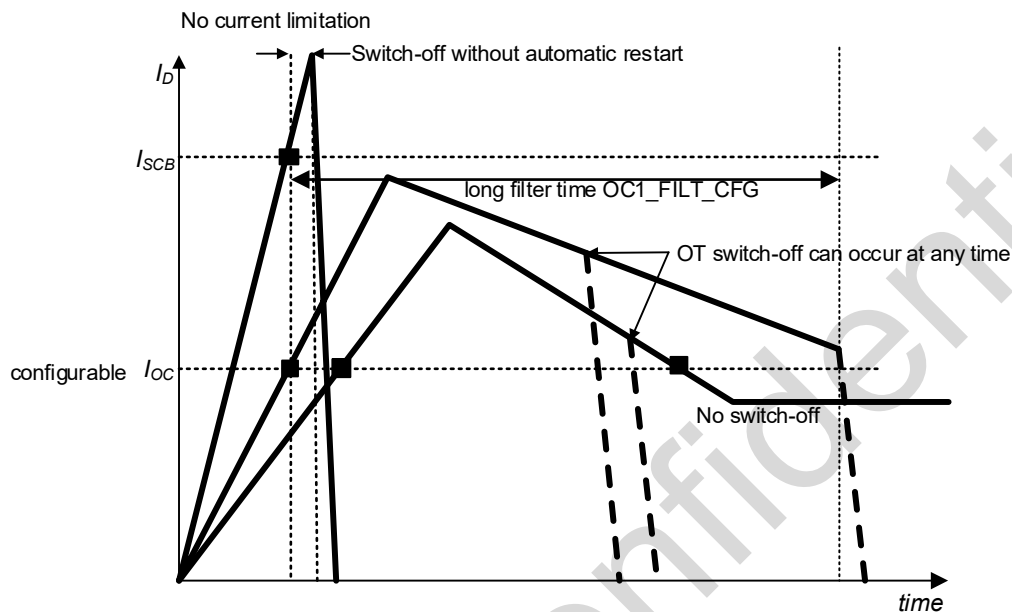


Figure 5-1 Overcurrent and short circuit to battery protection

The fault conditions SCG and OL will not be stored until an integrated filtering time, $t_{d(fault)}$, has expired. An additional blanking time, $t_{b(fault)}$, can be configured in addition to the filter time. The blanking time, $t_{b(fault)}$, can only be configured while output enable bit, OUT_EN, is low in the SPI register.

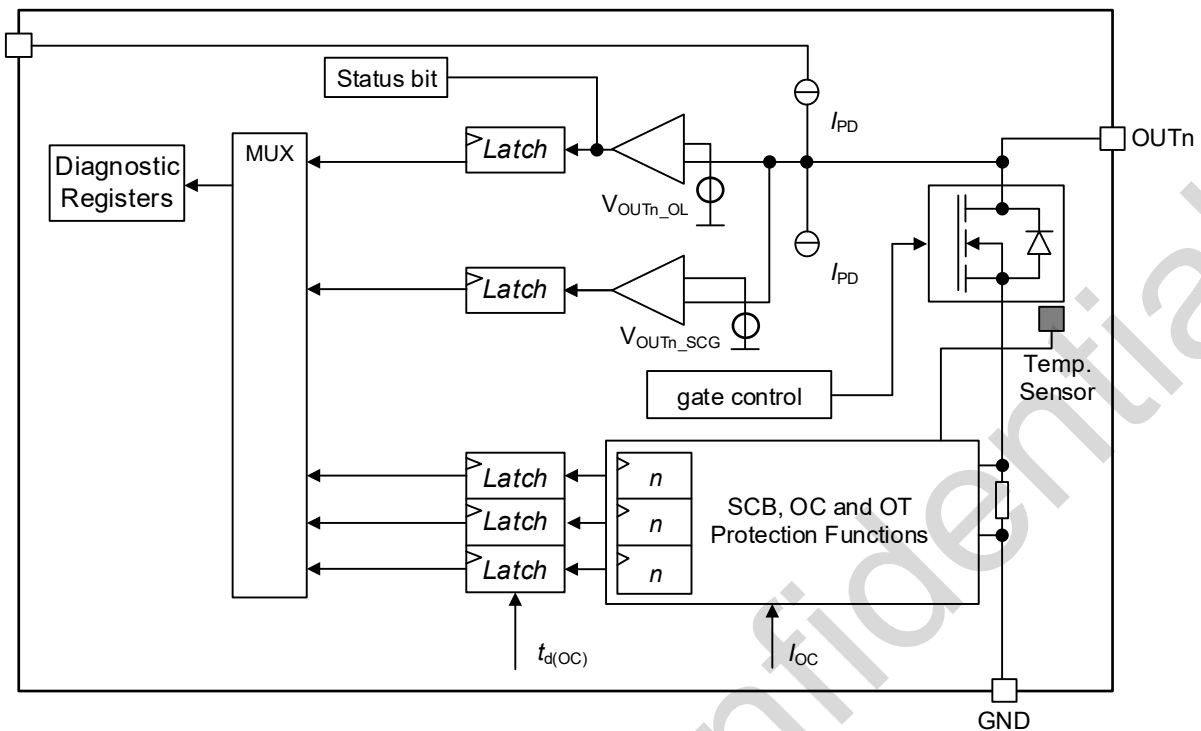


Figure 5-2 Diagnostic functions (overview only)

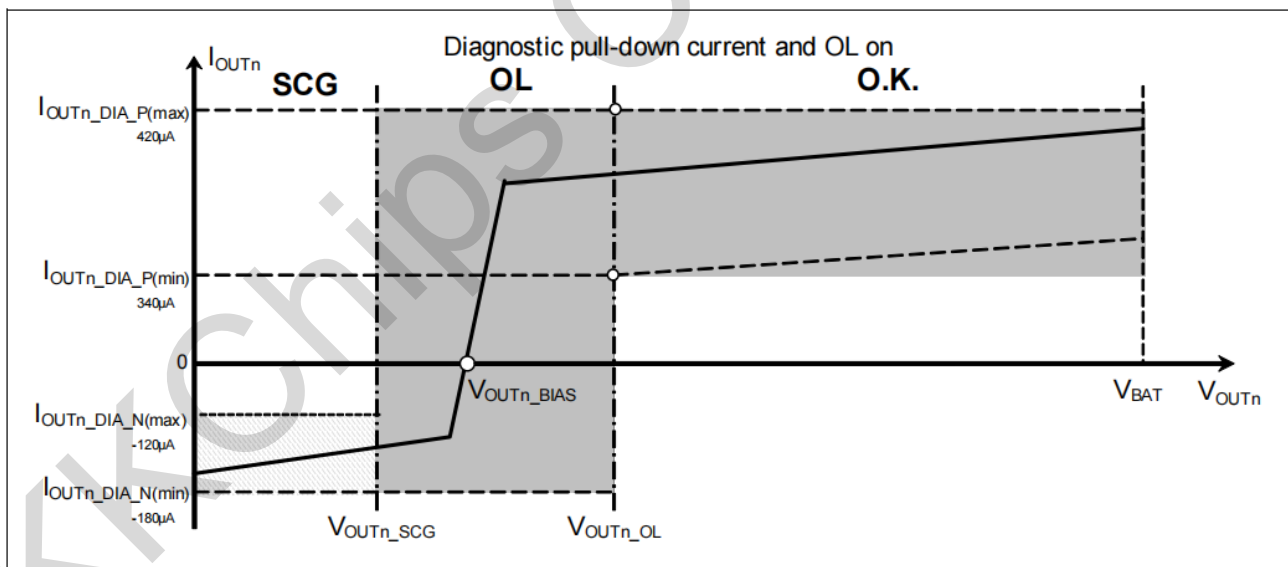


Figure 5-3 SCG and OL diagnostic function (overview only)

$V_{DD} = 4.5 \text{ V to } 5.5 \text{ V}$, $V_{IO} = 3 \text{ V to } 5.5 \text{ V}$, $T_j = -40^\circ\text{C to } +150^\circ\text{C}$, (unless otherwise specified)
all voltages with respect to ground, positive current flowing into pin (unless for pin SO)

Table 5-5 Diagnostic functions electrical characteristics

Symbol	Parameter	Condition	Min	Typ	Max	Unit	Param Num
$V_{OUTn-OL}$	Open load detection voltage	–	$0.6 V_{DD} - 0.2$	–	$0.6 V_{DD} + 0.2$	V	P_5.5.1
I_{PD}	Diagnostic pull-down current	$V_{OUTn} = 0.6 V_{DD}$ @ $T_j = -40^\circ\text{C to } +150^\circ\text{C}$	300	380	450	μA	P_5.5.2
I_{PU}	Diagnostic pull-up current	$V_{OUTn} = 0.4 V_{DD}$ @ $T_j = -40^\circ\text{C to } +150^\circ\text{C}$	-180	-150	-120	μA	P_5.5.3
$V_{OUTn-SCG}$	Short circuit to ground detection voltage	–	$0.4 V_{DD} - 0.2$	–	$0.4 V_{DD} + 0.2$	V	P_5.5.4
I_{SCB}	Short circuit to battery detection current	–	4.5	5	5.6	A	P_5.5.5
I_{SCB}	Short circuit to battery detection current in parallel mode	Outputs 1&2 or outputs 3&4 connected in parallel	9	10	11	A	P_5.5.6
$t_{d(fault)}$	Fault filtering time ¹⁾		0.015	0.02	0.025	ms	P_5.5.7
$t_{b(fault)}$	Fault blanking time ²⁾	configurable via SPI	0.16	0.2	0.24	ms	P_5.5.8
$t_{b(fault)}$	Fault blanking time	default value	0.4	0.5	0.60	ms	P_5.5.9
$t_{b(fault)}$	Fault blanking time	configurable via SPI	0.8	1	1.2	ms	P_5.5.10
$t_{b(fault)}$	Fault blanking time	configurable via SPI	1.6	2	2.4	ms	P_5.5.11
$t_{d(OC)}$	Overcurrent filtering time	default value	0.04	0.06	0.08	ms	P_5.5.12
$t_{d(OC)}$	Overcurrent filtering time	configurable via SPI	0.1	0.12	0.14	ms	P_5.5.13
$t_{d(OC)}$	Overcurrent filtering time	configurable via SPI	0.4	0.5	0.6	ms	P_5.5.14
$t_{d(OC)}$	Overcurrent filtering time	configurable via SPI	0.8	1	1.2	ms	P_5.5.15
$t_{d(OC)}$	Overcurrent filtering time	configurable via SPI	4	5	6	ms	P_5.5.16
$t_{d(OC)}$	Overcurrent filtering time	configurable via SPI	8	10	12	ms	P_5.5.17
$t_{d(OC)}$	Overcurrent filtering time	configurable via SPI	16	20	24	ms	P_5.5.18
$t_{d(OC)}$	Overcurrent filtering time	configurable via SPI	24	30	36	ms	P_5.5.19
I_{OC}	Overcurrent threshold	configurable via SPI	0.75	1	1.25	A	P_5.5.20
I_{OC}	Overcurrent threshold	default value	1.75	2	2.25	A	P_5.5.21

$V_{DD} = 4.5 \text{ V to } 5.5 \text{ V}$, $V_{IO} = 3 \text{ V to } 5.5 \text{ V}$, $T_j = -40^\circ\text{C to } +150^\circ\text{C}$, (unless otherwise specified)
all voltages with respect to ground, positive current flowing into pin (unless for pin SO)

Table 5-5 diagnostic functions (cont'd) electrical characteristics

Symbol	Parameter	Condition	Min.	Typ.	Max.	Unit	Param Num
I_{OC}	Overcurrent threshold	configurable via SPI	2.5	3	3.5	A	P_5.5.22
I_{OC}	Overcurrent threshold	configurable via SPI	3.5	4	4.5	A	P_5.5.23
I_{OC}	Overcurrent threshold in parallel mode	configurable via SPI, outputs 1&2 or outputs 3&4 connected in parallel	1.35	2	2.5	A	P_5.5.24
I_{OC}	Overcurrent threshold in parallel mode	default value, outputs 1&2 or outputs 3&4 connected in parallel	3.15	4	4.5	A	P_5.5.25
I_{OC}	Overcurrent threshold in parallel mode	configurable via SPI, outputs 1&2 or outputs 3&4 connected in parallel	4.5	6	7	A	P_5.5.26
I_{OC}	Overcurrent threshold in parallel mode	configurable via SPI, outputs 1&2 or outputs 3&4 connected in parallel	6.3	8	9	A	P_5.5.27
t_{OT}	Overttemperature filter time	—	2	3	4	μs	P_5.5.28
t_{SCB}	Short circuit to battery filter time	—	1.2	—	2	μs	P_5.5.29

1) $t_d(\text{fault})$ is the filter time for open load and short to ground diagnostic functions.

2) $t_d(\text{fault})$ is the blanking time for open load and short to ground diagnostic functions.

5.5.1 Output stage status

The output status of each channel can be directly obtained through the OUTx_STAT bit. This bit can be used to detect a fault condition where the channel is turned on via INx or SPI, but the power stage remains in the off state.

5.6 Communication watchdog

The HE9104C is using the watchdog principle to monitor the SPI communication. In case of no communication or continuous communication failures all outputs are disabled. In case of a faulty SPI frame the CWD timer does not retrigger and after the filter time the register CWD-TO is set and can be read as soon as the SPI is back to normal operation. The watchdog is active by default; however, it can be deactivated via a SPI command.

The watchdog starts to work as soon as the device has finished start-up and all blocks are released from reset. If these conditions are met, the watchdog timer t_{CWD} is started. Each correct SPI communication restarts the t_{CWD} timer. If no valid communication is received within timeout, the t_{CWD} timer will expire and disable all outputs. For re-enabling, one needs to clear the error and enable outputs via SPI. Outputs will not be enabled automatically by clearing the error.

The watchdog timer t_{CWD} is configurable via SPI. The watchdog timer t_{CWD} can only be configured while the output enable bit, OUT_EN, is low in the SPI register.

Following SPI communication issues are detected as failure by the watchdog:

- No communication
- Wrong commands
- Frames not equal to 16 clocks

$V_{DD} = 4.5 \text{ V to } 5.5 \text{ V}$, $V_{IO} = 3 \text{ V to } 5.5 \text{ V}$, $T_j = -40^\circ\text{C to } +150^\circ\text{C}$, (unless otherwise specified)
all voltages with respect to ground, positive current flowing into pin (unless for pin SO)

Table 5-6 Communication watchdog timeout configuration

Symbol	Parameter	Condition	Min.	Typ.	Max.	Unit	Param Num
t_{CWD0}	Communication watchdog timeout	configurable via SPI	20	25	30	ms	P_5.6.1
t_{CWD1}	Communication watchdog timeout	default value	40	50	60	ms	P_5.6.2
t_{CWD2}	Communication watchdog timeout	configurable via SPI	60	75	90	ms	P_5.6.3

6 16 bit SPI interface

The diagnostic and control interface is based on a serial peripheral interface (SPI).

The SPI is a full duplex synchronous serial slave interface, which uses four lines: SO, SI, SCK, CSN. Data is transferred by the lines SI and SO at the data rate given by SCK. The falling edge of CSN indicated the beginning of a data access. Data is sampled in on line SI at the falling edge of SCK and shifted out on line SO at the rising edge of SCK. Each access shall be terminated by a rising edge of CSN. A modulo 16 counter ensures that data is taken only, when a multiple of 16 bits has been transferred.

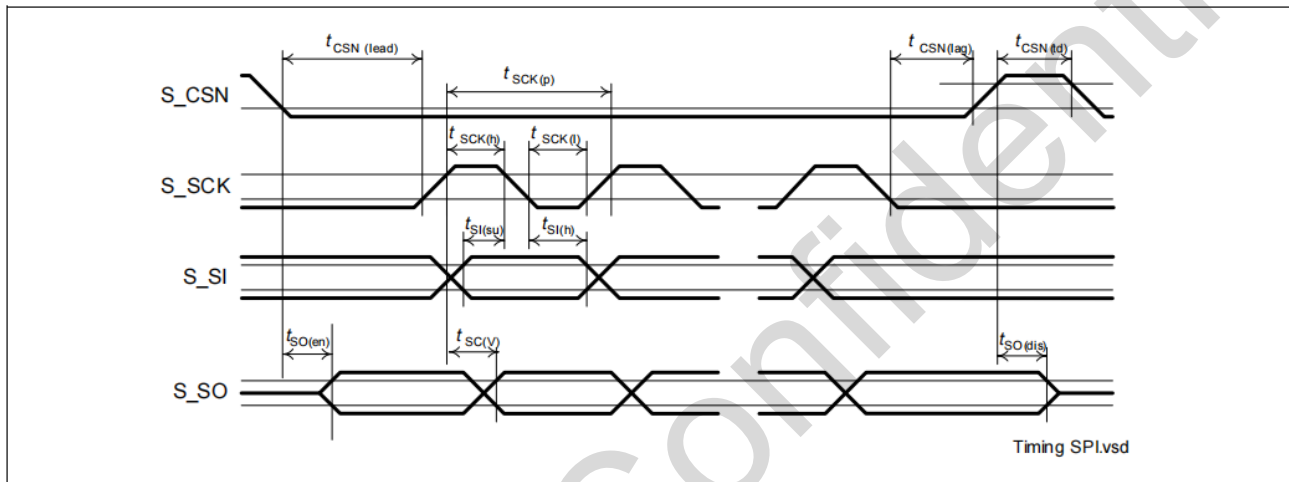


Figure 6-1 SPI timing

6.1 Electrical characteristics 16 bit SPI interface

$V_{DD} = 4.5\text{ V to }5.5\text{ V}$, $V_{IO} = 3\text{ V to }5.5\text{ V}$, $T_J = -40^{\circ}\text{C to }+150^{\circ}\text{C}$

all voltages with respect to ground, positive current flowing into pin (unless otherwise specified)

Table 6-1 16 bit SPI interface electrical characteristics

Symbol	Parameter	Condition	Min	Typ	Max	Unit	Param Num
Input characteristics (CSN, SCK, SI)							
V_{IO_CSNL} V_{IO_SCKL} V_{IO_SIL}	L level of pin CSN, SCK, SI	—	-0.3	—	1	V	P_6.1.1
V_{IO_CSNH} V_{IO_SCKH} V_{IO_SIH}	H level of pin CSN, SCK, SI	—	2	—	$V_{IO} + 0.3$	V	P_6.1.2
V_{IO_CSNhy} V_{IO_SCKhy} V_{IO_SIhy}	Hysteresis input pins	—	100	300	500	mV	P_6.1.3
Output characteristics (SO)							

HE9104C

Smart Quad Channel Powertrain Switch



V_{IO_SOL}	L level output voltage	$I_{IO_SO} = -2 \text{ mA}$	0	–	1	V	P_6.1.4
V_{IO_SOH}	H level output voltage	–	$V_{IO} - 0.5$	–	$V_{IO} + 0.3$	–	P_6.1.5
I_{IO_SOoff}	Output tristate leakage current	–	-10	–	10	μA	P_6.1.6
Input capacitance							
C_{in}	For CSN, SCK, SI and SO	-	-	6	8	pF	P_6.1.13
Timings							
f_{SCK}	Serial clock frequency ¹⁾	$C_L = 25 \text{ pF}$	0	–	8	MHz	P_6.1.7
$t_{SCK(P)}$	Serial clock period	–	125	–	–	ns	P_6.1.8
$t_{SCK(h)}$	Serial clock high time	–	50	–	–	ns	P_6.1.9
$t_{SCK(l)}$	Serial clock low time	–	50	–	–	ns	P_6.1.10
$t_{CSN(lead)}$	Enable lead time (falling CSN to rising SCK)	–	250	–	–	ns	P_6.1.11
$t_{CSN(lag)}$	Enable lag time (falling SCK to rising CSN)	–	250	–	–	ns	P_6.1.12
$t_{SI(su)}$	Data setup time (required time SI to falling SCK)	–	20	–	–	ns	P_6.1.14
$t_{SI(h)}$	Data hold time (falling SCK to SI)	–	20	–	–	ns	P_6.1.15
$t_{SO(en)}$	Output enable time (falling CSN to SO valid)	$C_L = 25 \text{ pF}$	–	–	200	ns	P_6.1.16
$t_{SO(dis)}$	Output disable time (rising CSN to SO tri-state)	$C_L = 25 \text{ pF}$	–	–	200	ns	P_6.1.17
$t_{SO(v)}$	Output data valid time with capacitive load	$C_L = 25 \text{ pF}$	–	–	100	ns	P_6.1.18
$t_{CSN(td)}$	Transfer delay time (rising CSN to falling CSN)	$C_L = 25 \text{ pF}$	1	-	100	us	P_6.1.19

1) Maximum SPI clock frequency in the application may be less depending on the load at the SO pin and the microcontroller SPI peripheral timing requirements.

6.2 SPI registers

The general SPI frame length is fixed at 16 bits. Bits 0 to 7 of each frame are used as data frame, bits 8 to 10 are used for address, bit 14 is the parity bit and bit 15 is used to specify a command as read or write. The parity bit is defined as:

$$B_{14} = (1 + b_{15} + \sum_{i=0}^{13} b_i) \bmod 2$$

(6.1)

MOSI

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R/W	Parity	0	0	Address				Data							

MISO

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R/W	Parity	Fault Communication	Fault Global	Address				Data							

IN MOSI and MISO a read is defined with a '0' and a write is defined with a '1'. Each MISO SPI frame reports the important system faults as Global or Communication faults in bit 12 and 13 as following:

- Global fault is asserted (not latching) when the general fault bit in Global_Status register is set.
- Communication fault is asserted (not latching) when one of the following faults are present:
 - Communication error (the same as COM_ERR in Global_Status register)
 - Communication watchdog timeout
 - Parity error

Besides, global status register stores the faults as following:

- General fault if at least one of the following faults are present:
 - Over-current
 - Over-temperature
 - Over-temperature during overcurrent
 - Short circuit to battery
 - Open load
 - Short circuit to ground
- Communication error:
 - No communication
 - Wrong command
 - Frames not equal to 16 bits
- Parity error
- Communication watchdog

Apart from the faults, global register also restores the enable latch signal (EN_Latch) and power on reset latch (POR_Latch) as following:

- EN_Latch: This bit has a reset value of '0'. After setting the OUT_EN bit this bit changes to '1'. This bit shows whether the output has been enabled (via SPI) at least once since the last clear.
- POR_Latch: This bit has a reset value of '1'. It can be changed to '0' via SPI. Any power on reset will set the bit back to 1. This can be used to check whether a power on reset has happened since the bit value was changed to '0'.

Table 6-2 Register Address Space

Module	Base Address	End Address	Note
apb	0H	1FH	–

Table 6-3 Register Overview

Register Short Name	Register Long Name	Offset Address	Param Num
CTRL	Output control register	00H	22
CFG	Configuration register	01H	24
OFF_DIAG_CFG	Off-state diagnostic configuration register	02H	25
ON_DIAG_CFG	On-state diagnostic configuration register	03H	26
DIAG_OUT_1_2_ON	On-state diagnostic result register OUT1 & OUT2	04H	27
DIAG_OUT_3_4_ON	On-state diagnostic result register OUT3 & OUT4	05H	28
DIAG_OFF	Off-state diagnostic result register	06H	29
GLOBAL_STATUS	Global device status register	07H	30
ICVID	IC Version ID	08H	31
CFG1	Configuration register1	09H	32

The registers are addressed wordwise.

Table 6-4 Register Overview

Bit type short name	Bit type description	Note
r	read	–
rw	read/write	–
rwc	read and clear on write	clear on write 0

Note: Except for OC_TH, all configurations can only be changed when the OUT_EN bit is cleared.

6.2.1 Registers

Output control register

CTRL	Offset	Reset value
Output control register	00H	00H

7	6	5	4	3	2	1	0
OUT4_ON_S	OUT4_ON_C	OUT3_ON_S	OUT3_ON_C	OUT2_ON_S	OUT2_ON_C	OUT1_ON_S	OUT1_ON_C
rw	rw	rw	rw	rw	rw	rw	rw

Field	Bits	Type	Description
OUT4_ON_S	7	rw	OUT4 SPI control bit (used if CFG.OUT4_DD = 0) 1 _D SET , Output 4 set 0 _D NO_ACTION , Output 4 no action Reset: 0 _B
OUT4_ON_C	6	rw	OUT4 SPI control bit (used if CFG.OUT4_DD = 0) 1 _D CLEAR , Output 4 clear 0 _D NO_ACTION , Output 4 no action Reset: 0 _B
OUT3_ON_S	5	rw	OUT3 SPI control bit (used if CFG.OUT3_DD = 0) 1 _D SET , Output 3 set 0 _D NO_ACTION , Output 3 no action Reset: 0 _B
OUT3_ON_C	4	rw	OUT3 SPI control bit (used if CFG.OUT3_DD = 0) 1 _D CLEAR , Output 3 clear 0 _D NO_ACTION , Output 3 no action Reset: 0 _B
OUT2_ON_S	3	rw	OUT2 SPI control bit (used if CFG.OUT2_DD = 0) 1 _D SET , Output 2 set 0 _D NO_ACTION , Output 2 no action Reset: 0 _B
OUT2_ON_C	2	rw	OUT2 SPI control bit (used if CFG.OUT2_DD = 0) 1 _D CLEAR , Output 2 clear 0 _D NO_ACTION , Output 2 no action Reset: 0 _B
OUT1_ON_S	1	rw	OUT1 SPI control bit (used if CFG.OUT1_DD = 0) 1 _D SET , Output 1 set 0 _D NO_ACTION , Output 1 no action Reset: 0 _B
OUT1_ON_C	0	rw	OUT1 SPI control bit (used if CFG.OUT1_DD = 0) 1 _D CLEAR , Output 1 clear 0 _D NO_ACTION , Output 1 no action Reset: 0 _B

HE9104C

Smart Quad Channel Powertrain Switch



Configuration register

CFG	Offset	Reset Value
Configuration register	01 _H	8F _H

7	6	5	4	3	2	1	0
CWD_TIME	OUT3_4_PAR	OUT1_2_PAR	OUT4_DD	OUT3_DD	OUT2_DD	OUT1_DD	
rw	rw	rw	rw	rw	rw	rw	rw

Field	Bits	Type	Description
CWD_TIME	7:6	rw	Communication watchdog timeout configuration 3 _D LONG , 75 ms 2 _D MEDIUM , 50 ms (default) 1 _D SHORT , 25 ms 0 _D DISABLED , Communication watchdog disabled Reset: 10 _B
OUT3_4_PAR	5	rw	OUT3-4 parallel mode 1 _D ENABLED , OUT3-4 parallel mode (controlled by IN3 or CTRL.OUT3_ON) 0 _D DISABLED , OUT3, OUT4 controlled separately (default) Reset: 0 _B
OUT1_2_PAR	4	rw	OUT1-2 parallel mode 1 _D ENABLED , OUT1-2 parallel mode (controlled by IN1 or CTRL.OUT1_ON) 0 _D DISABLED , OUT1, OUT2 controlled separately (default) Reset: 0 _B
OUT4_DD	3	rw	OUT4 direct drive mode 1 _D ENABLED , OUT4 controlled by IN4 (default) 0 _D DISABLED , OUT4 controlled by SPI (CTRL.OUT4_ON) Reset: 1 _B
OUT3_DD	2	rw	OUT3 direct drive mode 1 _D ENABLED , OUT3 controlled by IN3 (default) 0 _D DISABLED , OUT3 controlled by SPI (CTRL.OUT3_ON) Reset: 1 _B
OUT2_DD	1	rw	OUT2 direct drive mode 1 _D ENABLED , OUT2 controlled by IN2 (default) 0 _D DISABLED , OUT2 controlled by SPI (CTRL.OUT2_ON) Reset: 1 _B
OUT1_DD	0	rw	OUT1 direct drive mode 1 _D ENABLED , OUT1 controlled by IN1 (default) 0 _D DISABLED , OUT1 controlled by SPI (CTRL.OUT1_ON) Reset: 1 _B

HE9104C

Smart Quad Channel Powertrain Switch



Off-state diagnostic configuration register

OFF_DIAG_CFG	Offset	Reset Value
Off-state diagnostic configuration register	02 _H	1F _H

7	6	5	4	3	2	1	0
RES		DIAG_FILT_CFG		OUT4_DI AG_EN	OUT3_DI AG_EN	OUT2_DI AG_EN	OUT1_DI AG_EN
		rw		rw	rw	rw	rw

Field	Bits	Type	Description
DIAG_FILT_CFG	5:4	rw	Diagnostic filter time configuration 3 _D 2000_us , 2000 us 2 _D 1000_us , 1000 us 1 _D 500_us , 500 us (default) 0 _D 200_us , 200 us Reset: 01 _B
OUT4_DIAG_EN	3	rw	Enable diagnostic current OUT4 1 _D ON , Diagnostic current ON (default) 0 _D OFF , Diagnostic current OFF Reset: 1 _B
OUT3_DIAG_EN	2	rw	Enable diagnostic current OUT3 1 _D ON , Diagnostic current ON (default) 0 _D OFF , Diagnostic current OFF Reset: 1 _B
OUT2_DIAG_EN	1	rw	Enable diagnostic current OUT2 1 _D ON , Diagnostic current ON (default) 0 _D OFF , Diagnostic current OFF Reset: 1 _B
OUT1_DIAG_EN	0	rw	Enable diagnostic current OUT1 1 _D ON , Diagnostic current ON (default) 0 _D OFF , Diagnostic current OFF Reset: 1 _B

On-state diagnostic configuration register

ON_DIAG_CFG	Offset	Reset Value
On-status diagnostic configuration register	03 _H	01 _H
76543210 RESOC_FILT_CFGOC_TH		

Field	Bits	Type	Description
OC_FILT_CFG	4:2	rw	Overcurrent shut-down delay time (for all channels) 7 _D 30_ms, 30 ms 6 _D 20_ms, 20 ms 5 _D 10_ms, 10 ms 4 _D 5_ms, 5 ms 3 _D 1_ms, 1 ms 2 _D 500_us, 500 us 1 _D 120_us, 120 us 0 _D 60_us, 60 us (default) Reset: 000 _B
OC_TH	1:0	rw	Overcurrent shut-down threshold (for all channels) (d_oc_th) 3 _D 4000_mA, 4 A 2 _D 3000_mA, 3 A 1 _D 2000_mA, 2 A (default) 0 _D 1000_mA, 1 A Reset: 01 _B

On-state diagnostic result register OUT1 & OUT2

DIAG_OUT_1_2_ON	Offset	Reset Value
On-state diagnostic result register OUT1 & OUT2	04 _H	00 _H

7	6	5	4	3	2	1	0
OUT2_ST AT	OUT1_ST AT	DIAG_CH2_ON			DIAG_CH1_ON		
r	r	rwc			rwc		

Field	Bits	Type	Description
OUT2_STAT	7	r	Channel 2 output status 1 _D ON , Channel is ON 0 _D OFF , Channel is OFF Reset: 0 _B
OUT1_STAT	6	r	Channel 1 output status 1 _D ON , Channel is ON 0 _D OFF , Channel is OFF Reset: 0 _B
DIAG_CH2_ON	5:3	rwc	On-state diagnostic result register - Channel 2 7 _D UNUSED , unused combination 6 _D UNUSED , unused combination 5 _D OT , Overtemperature 4 _D OC_TIME , Overcurrent timeout 3 _D OC_OT , Overtemperature during overcurrent 2 _D SCB , Short to battery 1 _D NO_FAIL , no failure detected 0 _D UNKNOWN , no diagnosis done Reset: 000 _B
DIAG_CH1_ON	2:0	rwc	On-state diagnostic result register - Channel 1 7 _D UNUSED , unused combination 6 _D UNUSED , unused combination 5 _D OT , Overtemperature 4 _D OC_TIME , Overcurrent timeout 3 _D OC_OT , Overtemperature during overcurrent 2 _D SCB , Short to battery 1 _D NO_FAIL , no failure detected 0 _D UNKNOWN , no diagnosis done Reset: 000 _B

On-state diagnostic result register OUT1 & OUT2

DIAG_OUT_3_4_ON	Offset	Reset Value
On-state diagnostic result register OUT3 & OUT4	05 _H	00 _H

7	6	5	4	3	2	1	0
OUT4_ST AT	OUT3_ST AT	DIAG_CH4_ON			DIAG_CH3_ON		
r	r	rwc			rwc		

Field	Bits	Type	Description
OUT4_STAT	7	r	Channel 4 output status 1 _D ON , Channel is ON 0 _D OFF , Channel is OFF Reset: 0 _B
OUT3_STAT	6	r	Channel 3 output status 1 _D ON , Channel is ON 0 _D OFF , Channel is OFF Reset: 0 _B
DIAG_CH4_ON	5:3	rwc	On-state diagnostic result register - Channel 4 7 _D UNUSED , unused combination 6 _D UNUSED , unused combination 5 _D OT , Overtemperature 4 _D OC_TIME , Overcurrent timeout 3 _D OC_OT , Overtemperature during overcurrent 2 _D SCB , Short to battery 1 _D NO_FAIL , no failure detected 0 _D UNKNOWN , no diagnosis done Reset: 000 _B
DIAG_CH3_ON	2:0	rwc	On-state diagnostic result register - Channel 3 7 _D UNUSED , unused combination 6 _D UNUSED , unused combination 5 _D OT , Overtemperature 4 _D OC_TIME , Overcurrent timeout 3 _D OC_OT , Overtemperature during overcurrent 2 _D SCB , Short to battery 1 _D NO_FAIL , no failure detected 0 _D UNKNOWN , no diagnosis done Reset: 000 _B

Off-state diagnostic result register

DIAG_OFF	Offset	Reset Value
Off-state diagnostic result register	06 _H	00 _H

7	6	5	4	3	2	1	0
DIAG_CH4_OFF		DIAG_CH3_OFF		DIAG_CH2_OFF		DIAG_CH1_OFF	
rwc		rwc		rwc		rwc	

Field	Bits	Type	Description
DIAG_CH4_OFF	7:6	rwc	Off-state diagnostic result register Channel 4 3 _D SCG , Short to ground 2 _D OL , Open load 1 _D NO_FAIL , no failure detected 0 _D UNKNOWN , no diagnosis Done Reset: 00 _B
DIAG_CH3_OFF	5:4	rwc	Off-state diagnostic result register Channel 3 3 _D SCG , Short to ground 2 _D OL , Open load 1 _D NO_FAIL , no failure detected 0 _D UNKNOWN , no diagnosis done Reset: 00 _B
DIAG_CH2_OFF	3:2	rwc	Off-state diagnostic result register Channel 2 3 _D SCG , Short to ground 2 _D OL , Open load 1 _D NO_FAIL , no failure detected 0 _D UNKNOWN , no diagnosis done Reset: 00 _B
DIAG_CH1_OFF	1:0	rwc	Off-state diagnostic result register Channel 1 3 _D SCG , Short to ground 2 _D OL , Open load 1 _D NO_FAIL , no failure detected 0 _D UNKNOWN , no diagnosis done Reset: 00 _B

Global device status register

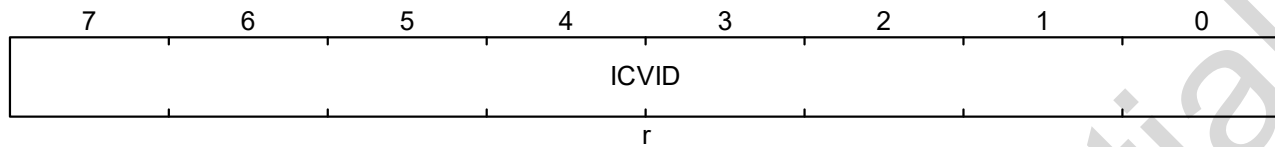
GLOBAL_STATUS	Offset	Reset Value
Global device status register	07 _H	01 _H

7	6	5	4	3	2	1	0
OUT_EN	SPARE	GEN_FAULT	COM_ERR	PAR_ERR	CWD_TO	EN_LATCH	POR_LATCH
rw	rw	rwc	rwc	rwc	rwc	rwc	rwc

Field	Bits	Type	Description
OUT_EN	7	rw	OUTx enable bit 1 _D ENABLED , Output switching enabled 0 _D DISABLED , Outputs disabled (default) Reset: 0 _B
SPARE	6	rw	Spare register for future use 1 _D GLOBAL_STATUS_SPARE_EN , 0 _D GLOBAL_STATUS_SPARE_DIS , (default) Reset: 0 _B
GEN_FAULT	5	rwc	General fault flag 1 _D ERR , At least one fault was detected 0 _D NO_ERR , No fault was detected Reset: 0 _B
COM_ERR	4	rwc	Communication Error Flag 1 _D ERR , At least one communication failure was detected 0 _D NO_ERR , No communication failure was detected Reset: 0 _B
PAR_ERR	3	rwc	Parity Error Flag 1 _D ERR , At least one parity error was detected 0 _D NO_ERR , No parity error was detected Reset: 0 _B
CWD_TO	2	rwc	Communication watchdog timeout 1 _D ERR , Communication watchdog timeout occurred 0 _D NO_ERR , No communication watchdog timeout (default) Reset: 0 _B
EN_LATCH	1	rwc	EN Latch 1 _D EN , Device was enabled since last read-out 0 _D NO_EN , Device was not enabled since last cleared Reset: 0 _B
POR_LATCH	0	rwc	Power-on reset latch 1 _D POR , Device was reset since last cleared 0 _D NO_POR , Device was not reset since last cleared Reset: 1 _B

IC Version ID

ICVID	Offset	Reset Value
IC Version ID	08 _H	B1 _H



Field	Bits	Type	Description
ICVID	7:0	r	IC Version ID 178 _D ICVID, Reset: B2 _H

Configuration register 1

CFG1	Offset	Reset Value
Configuration register 1	09 _H	00 _H

7	6	5	4	3	2	1	0
RES						SR	

Field	Bits	Type	Description
SR	1:0	rw	Turn on time and turn off time configuration 3 _D 2.5_us , 2.5us 2 _D 5_us , 5us 1 _D 10_us , 10us 0 _D 15_us , 15us Reset: 00 _B

7 Package outlines

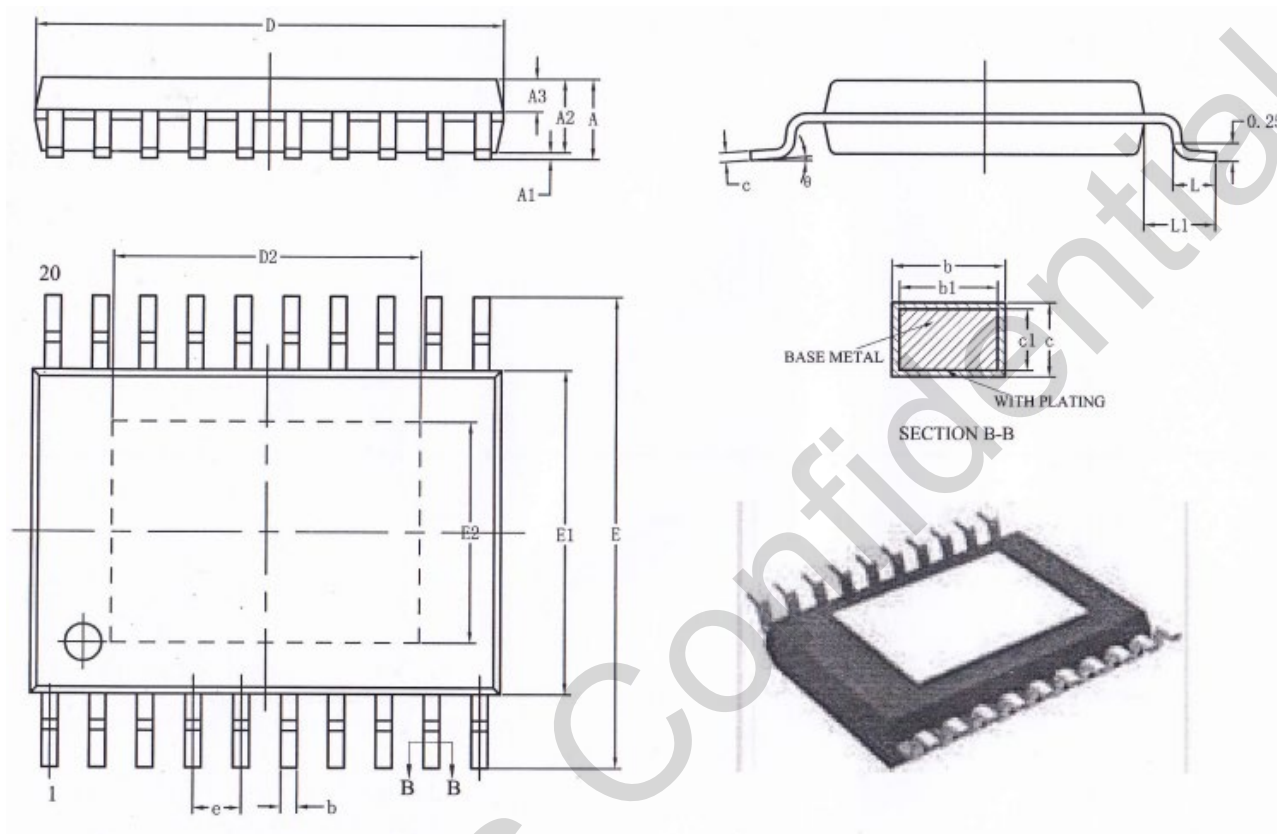


Figure 7-1 ETSSOP20 package dimensionw

Table 7-1 ETSSOP20L mechanical data

Symbol	Millimeter		
	Min	Nom	Max
A	-	-	1.20
A1	0.05	-	0.15
A2	0.80	1.00	1.05
A3	0.39	0.44	0.49
b	0.20	-	0.29

HE9104C

Smart Quad Channel Powertrain Switch



b1	0.19	0.22	0.25
c	0.13	-	0.18
c1	0.12	0.13	0.14
D	6.40	6.50	6.60
D2	4.10	4.20	4.30
E2	2.90	3.00	3.10
E1	4.30	4.40	4.50
E	6.20	6.40	6.60
e	0.65BSC		
L	0.45	0.60	0.75
L1	1.00REF		
θ	0	-	8°

8 Application information

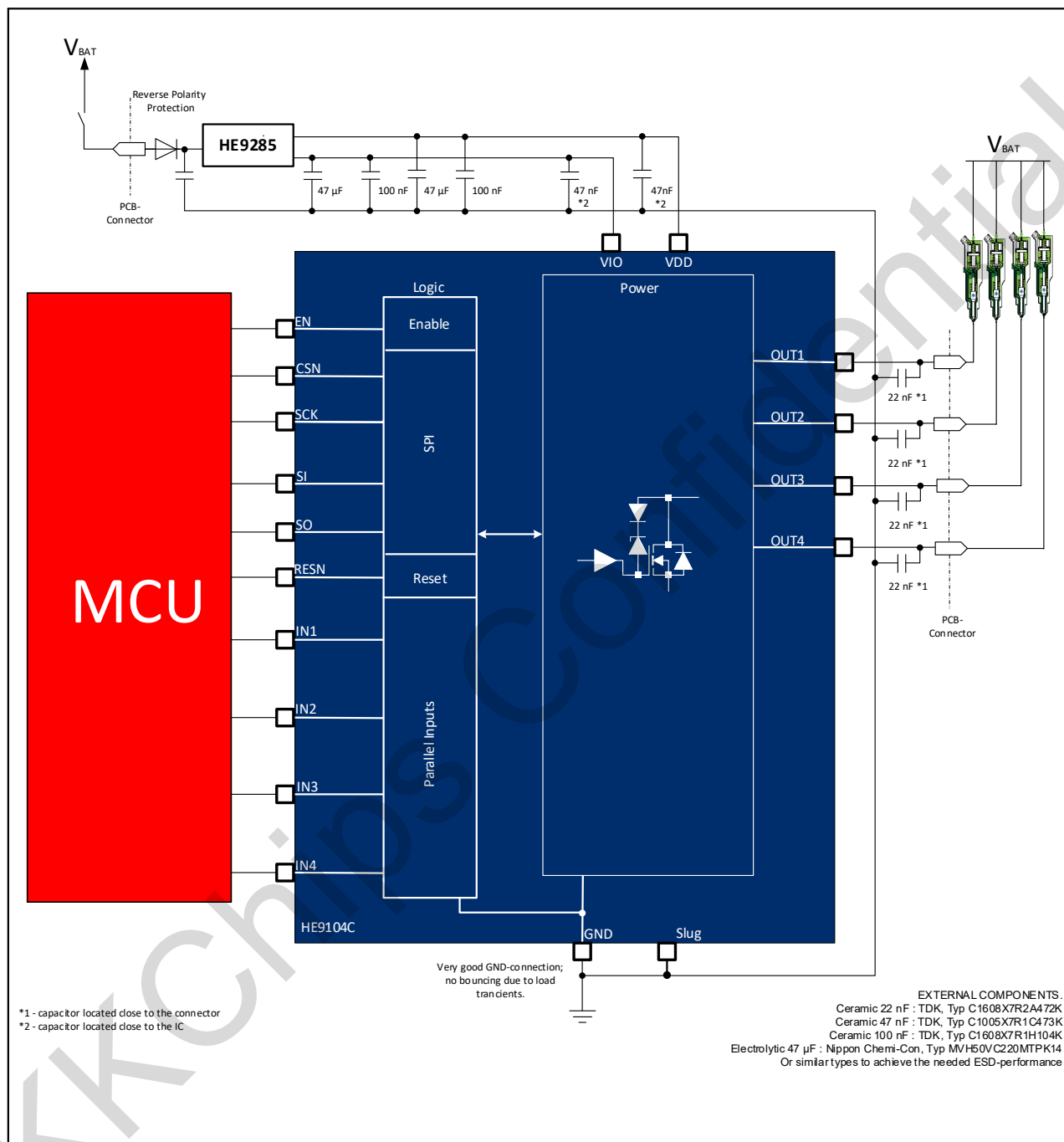


Figure 8-1 Multi port injection application diagram

9 Revision history

Table 9-1 Revision history

Version	Date	Changes
1.0	2025/11/10	1. First datasheet release.
1.1	2025/11/26	1. Revise the packaging specification from 4500 to 3000.
1.2	2026/5/25	1. Revise the incorrect description of the reset signal. 2. Change in AEC-Q100 qualification status.